News-Based Inflation Risk and Price Dynamics in Africa:

Evidence from a Panel VAR

CAPE Economic Research and Consulting

October 2026

Abstract

We construct the African Inflation Risk Index (AIRI) from newspaper reports on inflation-related risks and examine its relationship with subsequent consumer price inflation. The index combines economic relevance screening, inflation-specific classification, severity scores and measures of textual emphasis. It is calculated from 21.30 million retained article records covering 54 African economies during 2015–2025, with country-month scores divided by the total number of retained articles. AIRI can be matched to observed CPI inflation in 53 economies; monthly exchange-rate data are available for a smaller sample. In the 45-country monthly regression, lagged AIRI has a coefficient of 0.0355 (two-way clustered standard error 0.0104; $p = 0.0006$) after controlling for inflation persistence, exchange-rate movements, country effects and calendar-month effects. The estimated relationship varies across countries. AIRI is also 1.090 index points higher following months of double-digit inflation ($p = 0.0056$), conditional on recent inflation and newspaper coverage. The AIRI coefficient in the inflation equation is 0.0368 higher above the 10 percent inflation cutoff ($p = 0.0589$); this difference is significant at the 10 percent level, but not at 5 percent. Out-of-sample tests at four horizons show lower forecast errors in some countries, although improvements are not widespread. The paper provides a transparent measure of inflation-related news risk and assesses its relationship with observed inflation, without treating the index as a measure of inflation itself.

Keywords: inflation risk; economic news; Africa; panel dynamics; forecast evaluation; newspaper indicators.
JEL classifications: E31, E37, C23, C53, O55.

1 Introduction

Monthly inflation statistics are released after the prices they measure have been collected. Between releases, central banks and other economic analysts must assess whether price

pressures are building or receding. Domestic newspapers report on food and fuel prices, exchange-rate changes, transport costs, supply interruptions and shortages. Some reports describe price increases already under way; others concern developments likely to affect prices later. This paper asks whether the inflation risks recorded in those reports help explain subsequent inflation after accounting for past inflation and observed macroeconomic conditions. Our approach builds on research using newspaper text to measure economic uncertainty and sentiment ([7, 45]) and on studies of news-based inflation information ([8, 4]).

We study the African Inflation Risk Index (AIRI), derived from the inflation component of the African Economic Risk Index (AERI). AERI classifies economic news into eight risk areas. For AIRI, only articles assigned to the inflation category contribute to the numerator, and each receives the severity and textual-emphasis weights specified by the broader framework. We divide the resulting score by all retained newspaper articles for that country and month. AIRI therefore rises when inflation-related coverage becomes more frequent, more severe in its wording or more prominent within the newspaper corpus. It is an index of newspaper risk reporting, not a percentage inflation rate; the levels of the two measures have no direct numerical interpretation in common.

The scoring method separates three features of the news record: the incidence of inflation-related articles, the severity of their language and their textual prominence. It applies predetermined rules rather than fitting weights to the observed inflation series. That distinction matters when an index is later used to study macroeconomic outcomes ([24]). Newspaper archives can provide a regular source of information across countries with uneven statistical coverage, but the resulting measure also reflects which newspapers enter the archive and what those outlets choose to report. We therefore examine the statistical relationship between AIRI and inflation without assuming that newspaper coverage changes prices. Models of delayed information transmission and inflation attention motivate that relationship; the current data do not directly observe household expectations or attention ([32, 17, 46]).

The analysis covers January 2015 to December 2025. The archive contains 21,303,573 article records across 54 country series. We match AIRI to monthly CPI inflation for 53 countries, giving 6,912 country-month observations. Exchange-rate data are available for 46 of those countries; 45 meet the history and completeness requirements of the four-horizon forecast comparison. Sample sizes therefore differ across the descriptive, regression and forecast exercises. The source workbook also contains observed quarterly real GDP growth, which we leave out of the monthly regressions rather than assign to months without directly observed GDP data.

We estimate separate country autoregressions and a monthly panel regression with

country and calendar-month effects. Standard errors for the latter are clustered by country and month to allow for dependence within each dimension. We then estimate recursive forecasts separately for each country using three models: lagged inflation alone, lagged inflation with exchange-rate depreciation, and the same variables with AIRI. Separate forecasting equations allow the coefficients to differ across economies.

The results indicate three main patterns. AIRI and CPI inflation are positively correlated within 40 of the 51 countries with sufficiently long and varying series. Lagged AIRI also has a positive coefficient in the monthly inflation regression after the other controls are included. Finally, AIRI readings are higher after double-digit inflation, and the estimated association with subsequent inflation is stronger in those observations. The difference between the two inflation coefficients is significant at 10 percent ($p = 0.0589$), though not at 5 percent. Forecast accuracy improves in some countries and at some horizons, but the gains do not extend across the panel.

Section 2 explains the news measure and the data. Section 3 sets out the estimation and forecast comparisons. Section 4 reports the results, and Section 5 concludes. The accompanying supplementary information contains the more detailed country and forecasting tables.

2 Related Literature and Economic Motivation

2.1 Inflation information and the timing of official statistics

Official consumer price indices measure changes in the prices of goods and services after transactions have taken place and the statistical agency has collected and processed the underlying observations. An index available before the next statistical release must therefore draw on information not yet incorporated into the published figure. That information might concern current retail prices, wages, exchange rates, the cost of imported goods, shortages or changes in distribution costs. It can also concern expectations about these developments. The distinction matters: a news report of an existing price increase may help predict the next published CPI observation without predicting a new inflation shock.

A distinction should also be drawn between alternative high-frequency data sources. Online retail prices can document transactions or offered prices much closer to the moment of sale ([13]), whereas Internet search activity measures interest in a topic rather than the topic's economic outcome ([14]). The World Uncertainty Index uses standardized country reports rather than this domestic newspaper architecture ([2]). Newspaper reports occupy another position: they describe prices, supply conditions and prospective decisions in

narrative form. They can consequently contain both observed and anticipated inflation-related developments, but are not a direct replacement for either retail price records or inflation expectation surveys.

The economic literature gives several reasons to expect information to reach prices and measured inflation at different times. In sticky-information models, firms and households do not update their information sets continuously ([32]). Rational-inattention models likewise allow the allocation of attention to vary with the perceived importance of a development ([47]). Surveys and research on the expectations of households and firms provide direct evidence of information rigidity in expectation formation ([50]) ([16, 17]). None of these arguments, by itself, implies that a newspaper index will improve a point forecast. They suggest that the timing and content of information deserve empirical attention.

Media reports have several roles in this setting. Journalists can document developments that are already affecting prices; news outlets can also report prospective changes in administered prices, tariffs, fuel costs and food availability. Newspaper coverage may influence what the public notices, but the index developed here records published language rather than readers' attention or inflation expectations. We distinguish these objects throughout the analysis. An association between AIRI and future CPI inflation is consistent with useful information arriving in newspapers before it is fully reflected in CPI. It is also consistent with some articles responding to price pressures that the published index has not yet captured.

The classical monetary literature established expectations and monetary policy as central elements of inflation dynamics ([22, 42]). Subsequent work has emphasized persistence, the importance of expectations and the difficulty of distinguishing changing shocks from changing responses to them ([33]). Experimental evidence shows that information about inflation can affect household choices ([18]), while reviews of central-bank communication discuss both its potential and its limitations for informing the public ([10]). These studies motivate the measurement of the information environment; they do not establish a causal transmission channel for AIRI.

Real-time work has shown that the date on which an economic series is observed can matter as much as the final historical value ([36]). Our retrospective exercises do not use archived publication-time vintages. We impose chronological ordering on forecast targets and regressors and assess the results as dated historical forecasts; we do not treat the exercise as a reconstruction of the complete information set held by a central bank at each date.

2.2 Newspaper-based measurement and inflation

Baker, Bloom and Davis ([7]) established an influential template for producing an economic uncertainty index from newspaper language. Their measure relies on transparent dictionary restrictions and is checked against other indicators. Subsequent work has used text to measure general news sentiment ([45]), economic policy uncertainty in individual countries ([25, 44]) and systemic economic narratives ([35]). These contributions show both the promise and the main difficulty of dictionary-based measures: a transparent rule can be applied consistently, but the result depends on the vocabulary, sources and text-cleaning decisions that define the rule.

The contribution of AIRI is narrower than that of a general uncertainty index. It is intended to measure newspaper discussion of inflation-related economic risks rather than an undifferentiated change in economic tone. The relevant distinction is not merely terminological. Reports about government financing, commercial banking, weather damage and exchange-rate depreciation may overlap with inflation risk, but they do not necessarily refer to rising consumer prices. A multi-pillar classification allows those risks to be recorded separately even when an article mentions more than one.

There is an existing literature on inflation measurement from digital information. Angelico and coauthors ([4]) examine inflation expectations derived from social-media material; Larsen, Thorsrud and Zhulanova ([30]) study inflation-related news information; and Barbaglia, Consoli and Manzan ([8]) consider forecasting with economic news. Text-based measures can contain inflation-relevant information, but the incremental value depends on the country, the topic, the outcome and the comparison model. The construction of an index and the demonstration that it is a consistently better forecast are distinct empirical tasks.

Other text applications provide useful contrasts. Newspaper tone has been associated with market developments ([49, 23]), and economic-news datasets have been used to predict macroeconomic indicators ([21, 26]). Machine-learning approaches to inflation prediction can be more flexible than lexicon rules but require separate consideration of model complexity and stability ([5]). AIRI makes a different design choice: it uses explicit, predetermined article scoring so that its meaning does not depend on a fitted model of observed inflation.

The general text-as-data literature emphasizes measurement validity, document selection and researcher decisions about dictionaries ([24]). That guidance has particular force when the distribution of newspapers differs sharply across countries. A count of inflation-related documents may grow because inflation becomes more salient, because the archive adds a newspaper, or because a source changes its publication volume. A volume-normalized

measure reduces one mechanical form of scale dependence, but does not remove changes in the composition of the underlying sources. Our index construction and coverage analysis are designed around that distinction.

2.3 Attention and double-digit inflation

The salience of inflation varies with economic conditions. Households may tolerate small, familiar price movements without revising expectations, while persistent or unusually large increases make prices a more frequent subject of discussion. Bracha and Tang ([11]) document a relationship between the level of inflation and attention. Korenok, Munro and Chen ([29]) examine attention thresholds using several external measures. Their research motivates testing whether inflation-related newspaper risk scores differ between inflation environments. It does not supply a universal threshold suitable for every country.

We distinguish two propositions that are often combined loosely in discussions of inflation attention. The *news-reporting proposition* is that measured inflation-related newspaper risk differs when inflation is high. This can be tested using AIRI as the outcome. The *inflation-association proposition* is that the coefficient linking lagged AIRI to subsequent CPI inflation differs by inflation regime. This can be tested with an interaction term. Neither test on its own establishes whether households pay more attention, whether firms revise prices in response to news, or whether the amount of inflation-related coverage causes the change in expectations.

The present analysis uses a threshold of 10 percent year-on-year inflation as an interpretable description of double-digit inflation, not as an estimated psychological attention threshold. We report the coefficient difference and its exact significance level and retain other threshold results in supporting tables. A different threshold would change the number and composition of observations assigned to each regime. Moreover, a high-inflation regime can coincide with exchange-rate instability, monetary disruptions and common global supply shocks. Country and calendar-month effects, inflation lags and exchange-rate controls help separate some of these influences, but cannot convert the reduced-form relationship into a behavioral causal estimate.

2.4 African inflation dynamics and panel heterogeneity

African inflation experiences differ in their exposure to imported food and energy, exchange-rate depreciation, agricultural output shocks and monetary conditions. Barnichon and Peiris ([9]), Durevall and Sjö ([20]), and Nguyen and coauthors ([34]) document different aspects of these mechanisms. Across countries, the same nominal depreciation can be

followed by quite different price responses because import shares, administered prices and exchange-rate arrangements vary. In addition, reporting practices and the depth of newspaper archives differ. A common newspaper-risk coefficient should therefore be treated as a summary, not a description of identical national relationships.

Macroeconomic data coverage also presents a research difficulty distinct from the newspaper classification problem. Historical official series are uneven across African economies and may be revised as statistical systems and national accounts are updated ([28]). Exchange-rate pass-through differs with monetary arrangements and economic structure ([43]); the role of food inflation differs with production and import exposure ([3]). These features support a careful treatment of national variation instead of interpreting the sample as an economically homogeneous block.

These differences motivate reporting country-specific results rather than relying exclusively on pooled estimates. The Mean Group principle developed by Pesaran and Smith ([40]) is relevant when dynamic coefficients vary across units. Panel VARs offer a way to organize multiple jointly evolving series ([31, 12]); the reduced-form coefficients must nevertheless be distinguished from structural responses. In particular, a Cholesky ordering does not identify a causal newspaper shock without credible exclusion restrictions and an economic model of contemporaneous interactions.

Cross-sectional dependence is an additional issue. A food-price shock affecting several countries can change CPI inflation and newspaper coverage at the same time. Standard inference that treats national observations as independent can be too optimistic. The common-factor approach of Pesaran ([38]) provides one econometric response; the empirical analysis here includes calendar-month effects and clusters its reported standard errors by both country and month. These choices permit common calendar disturbances and within-country persistence without requiring a homogeneous VAR across all countries.

2.5 Research questions and contribution

Against this background, the paper addresses four questions. First, can the AERI newspaper architecture yield a well-defined country-month measure of reported inflation risk that is comparable in interpretation, though not necessarily identical in source composition, across African economies? Second, does lagged AIRI retain an association with subsequent CPI inflation when inflation persistence, observed exchange-rate movements and common calendar shocks are controlled for? Third, do the level of newspaper risk and the conditional AIRI coefficient differ in double-digit inflation environments? Fourth, does the index improve recursive inflation forecasts over models based on inflation history and exchange-rate movements, and is that result consistent across countries?

Our contribution is principally one of measurement and empirical evaluation. The index uses a predetermined article-scoring architecture that treats the occurrence of inflation-related news, severity language and textual prominence as distinct inputs. We apply it to a broad collection of domestic newspaper records and then examine country-specific and pooled inflation relationships using observed macroeconomic series. We do not claim to have invented panel VARs, clustered inference, threshold regressions or forecast-comparison methods. The distinct research object is AIRI and the evidence on how the economic information in that measure varies across countries and inflation conditions.

3 AIRI Measurement and Data

3.1 The inflation-risk index

Write $\mathcal{A}_{ct}$ for the set of retained newspaper articles from country c in month t, and let $N_{ct} = |\mathcal{A}_{ct}|$. The indicator R_i equals one when article i passes the economic-relevance screen; $I_{i,\pi}$ equals one when it also matches the inflation-risk dictionary. Severity S_i ranges from 0 to 4. Articles that do not pass the relevance screen receive zero, while relevant articles receive the highest severity tier activated by their text. The textual-emphasis weight is

$$W_i = \min\{2,\, 1 + 0.5T_i + 0.2L_i + 0.3M_i\}, \tag{1}$$

Here T_i marks a risk-related phrase in the headline, L_i marks an article body of at least 1,200 normalised characters, and M_i marks an article assigned to two or more economic-risk categories. All three indicators are binary. The weight reflects the prominence and breadth of risk language in an article; it does not measure the accuracy of the report or the reliability of its publisher. The country-month inflation-risk score is

$$\text{AIRI}_{ct} = \frac{100}{N_{ct}} \sum_{i \in \mathcal{A}_{ct}} R_i S_i W_i I_{i,\pi}. \tag{2}$$

Articles that fail the relevance screen contribute nothing to the numerator, although they remain part of the total archive count. An article assigned to several risk areas contributes to AIRI only if it also activates the inflation dictionary. Given $R_i I_{i,\pi} \leq 1$, $S_i \leq 4$ and $W_i \leq 2$, equation (2) has a theoretical upper bound of 800. This is a mathematical bound, not the scale normally observed in the data. Among the country-month observations matched to CPI, AIRI ranges from 0 to 54.00.[1]

[1]The separate African Risk Score uses a divisor of 200 to construct a bounded score. AIRI does not use that divisor; its regression coefficient must be interpreted in AIRI index points.

Two months with the same number of inflation-related articles need not have the same AIRI score. Reporting that uses stronger severity terms or carries more headline emphasis receives a larger weight. The total number of articles also affects the score: AIRI can fall if the overall archive expands while the volume and composition of inflation-related reporting remain unchanged. The measure describes the content of the newspapers observed in our archive. It does not estimate the share of the population exposed to inflation news, and changes in the source roster can affect comparisons over time.

The country-month files retain the total article count, economically relevant article count, share of economic reporting and scores for all eight risk areas. These variables permit comparisons of AIRI with observed inflation and checks on news coverage. They do not, on their own, show whether every underlying article was classified correctly; article-level classification accuracy is a distinct question from the macroeconomic relationship examined here.

3.2 From newspaper articles to a country-month observation

The basic unit of the index is an article, not a search query or the output of a forecasting model. Each retained record is associated with a source, a country and a publication month. The procedure is applied separately to each country-month archive. Articles that are not classified as economically relevant remain in the denominator, while their contribution to the scored inflation-risk numerator is zero. This design measures the prominence of inflation-related risk within the retained newspaper corpus, rather than the average severity among only those documents selected for inflation discussion.

This choice has an interpretation that is sometimes missed. Suppose two countries each have one hundred inflation-risk reports carrying an identical average article score. The country with twice as many retained articles will receive half the AIRI score, other things equal. That is not a coding error: a fixed number of inflation reports represents a smaller share of a larger newspaper archive. The rule mitigates the mechanical effect of the scale of the media corpus. It does not make the two newspaper systems representative of identical populations of information producers or readers.

The scoring architecture first identifies economically relevant material through a dictionary applied to the available title and text. The inflation pillar is one among eight broader risk categories. An article may appear in several risk pillars; for example, a report on exchange-rate depreciation and the rising cost of imported food can register both external-sector and inflation risk. This multi-label design prevents the researcher's choice of a single exclusive category from suppressing overlapping risk narratives. It also means that a sum of all pillar scores should not be interpreted as a count of distinct inflation or

economic articles.

Severity is assigned on an ordered scale from zero to four, according to the strongest matching severity category. The index therefore differs from simple inflation-news frequency by allowing two inflation articles to make different contributions. Textual prominence is measured through fixed increments for a qualifying headline cue, sufficient body length and activation of several risk pillars. The upper bound on the weight prevents these features from giving any one document an unlimited contribution. These are classification and salience conventions; they should not be read as estimates of a newspaper's journalistic reliability. The reported "confidence" fields in the production dataset are retained for consistency with the original code, while the paper interprets the weights by the textual features they actually measure.

For illustration, consider two relevant inflation articles, both with severity tier three. An article with no additional cues receives a score of three. A second article with a qualifying headline cue and sufficiently long text receives a larger score under the stated weighting formula. The difference is a mechanical consequence of observable textual features, not an estimate of the economic damage from the event described. Similarly, an article mentioning inflation as well as foreign-exchange and debt risks can receive a multi-pillar increment. The formula preserves these distinctions at the aggregation stage.

3.3 Index units, non-negativity and normalization

An article's score is bounded above by eight under the declared rules. Dividing the total score of inflation-tagged articles by the complete article count and multiplying by one hundred yields an index expressed in scored inflation-risk units per one hundred retained articles. Its theoretical bound is zero to eight hundred if all retained articles activate the maximum score. The realized series occupies a far narrower range. It is therefore misleading to describe the production AIRI series as a 0–200 index without demonstrating another transformation. The separate African Risk Score in the broader AERI system employs different scaling conventions; it is not interchangeable with the inflation pillar used here.

The level of AIRI should not be compared numerically with CPI inflation. A value of eight AIRI points does not represent eight percent inflation, and the difference between the two series is not an inflation forecast error. Standardized plots can compare their movement through time, while correlations and regressions assess associations after accounting for their separate units. A coefficient on AIRI in an inflation equation is expressed as percentage points of annual CPI inflation per one AIRI index point, conditional on the other regressors in that equation.

There are two relevant normalization questions. First, does dividing by retained article counts address the fact that some countries have much larger newspaper archives than others? It removes proportional changes in all article counts when the composition of reporting is unchanged. Second, does it eliminate the effect of adding or removing newspapers with different editorial priorities? No. If a newly added newspaper publishes comparatively little inflation-related reporting, the measured share can decline even if the existing sources publish exactly the same content. Both statements are consequences of the definition and should be kept distinct.

3.4 Source coverage and observations with little information

The archived AERI production files include country-month article totals, economically relevant counts, the share of economically relevant articles, average severity among economically relevant articles and the eight pillar scores. These fields permit checks on volume and broad economic-news composition. The article-level dictionary matches cannot, however, be recovered uniquely from country-month totals. The same aggregate AIRI value could arise from many combinations of article counts, severity tiers and textual weights. A reproducible calculation from raw articles therefore requires the original source registry, text-processing rules and classification dictionaries in addition to the monthly export.

For 2015–2025, the 54-country newspaper archive contains 7,127 country-month records. In 639 of those months, fewer than one hundred articles were retained. Such months may contain genuine reporting, but a small change in article composition can have a larger effect on the index than in months with several thousand documents. We retain these observations in the primary CPI analysis and report a sensitivity comparison after excluding low-volume months. Missing country-month records are not assigned a zero value. An observed zero means that the retained news records yielded no positive inflation-risk score under the coding rules; a missing value indicates that a usable country-month index observation was not present.

This distinction is especially important for countries with long stretches of zeros. In Equatorial Guinea and Guinea-Bissau the available AIRI series is constant at zero, and the evaluation-origin period for Mauritania also contains no variation in AIRI. A constant predictor cannot independently explain changes in inflation within that country, and the country-specific forecast results need to be interpreted accordingly. We neither silently drop these countries from the source-coverage totals nor treat their zero scores as proof that inflation risk was absent in the economy.

3.5 Language, terminology and the scope of the evidence

A multilingual news archive poses issues beyond translating individual inflation words. Economic terminology, monetary institutions, currency conventions and the use of metaphor vary across languages and countries. A dictionary must identify which concepts activate economic relevance, which activate the inflation pillar and which severity terms alter the score. The source registry and dictionaries are integral parts of the index production process. Country-month outputs alone do not show the language-specific recall and precision of article classifications.

The broader archive was assembled from English-, French-, Arabic- and Portuguese-language news sources. That establishes the linguistic scope of collection; it does not, without article-level tests, prove that inflation terms are equally well detected in every language. The paper accordingly describes AIRI as a multilingual newspaper-based index and separates its reproducible aggregation formula from the still distinct task of validating individual text classifications.

A fully transparent text measurement release has two layers. One is the analytical panel: country, month, index score, article totals, macroeconomic outcomes and the code for the reported regressions. The second is the upstream measurement archive: source lists, date-processing rules, text deduplication rules, language dictionaries and matched article features. The analyses in this paper can be reproduced from the first layer. Reconstructing the index article by article depends on the second. This distinction is presented here as a description of the data architecture rather than as a substitute for measurement validation.

3.6 A simple benchmark for the measurement design

The conceptual comparator to AIRI is the share of inflation-related articles among all retained articles. Under uniform article weights, that quantity depends only on how often inflation is discussed. AIRI adds the ordered severity score and textual features, and therefore can change even when the share of inflation articles is unchanged. The contrast defines what the severity-weighted framework attempts to measure: not only the prevalence of inflation discussion, but the intensity of the reported threats under a fixed scoring rule.

The monthly aggregate files do not separately identify the number of inflation-tagged documents. It would therefore be incorrect to construct a supposedly unweighted inflation-news frequency measure by dividing total economically relevant articles by all articles. Economic relevance includes numerous topics outside inflation. We retain the distinction between these categories rather than report a misleading comparison. The analysis

evaluates the full AIRI series and, as coverage checks, includes measures of overall economic-news share and average severity among economically relevant material. A direct frequency-versus-weighted inflation comparison requires the separate inflation-tagged article counts.

3.7 News coverage, CPI and exchange rates

For 2015–2025, the archive records 7,127 country-month observations. In 639 of these months, fewer than 100 articles were retained; 6,488 months meet or exceed that count. Our main CPI regressions include the low-volume months, and a separate estimate excludes months with fewer than 100 articles. We treat missing archive months as missing observations, not as zero inflation risk. Before merging with the macroeconomic data, we standardise country names, including Cabo Verde/Cape Verde and alternative names for the Republic of the Congo and the Democratic Republic of the Congo.

We take headline CPI inflation and nominal exchange rates from the monthly worksheets in *Selected Macro Indicators.xlsx*. Exchange rates are measured as units of local currency per US dollar. For consecutive months with positive exchange-rate observations, depreciation is $100[\log(e_{ct}) - \log(e_{c,t-1})]$; a positive value denotes a depreciation. Table 1 shows the coverage after these variables are matched. Real GDP growth is observed quarterly in the available workbook, so it does not enter our monthly specifications. We also exclude money growth because the verified data do not contain a complete corresponding monthly M2 panel.

Table 1: Archive and estimation samples, January 2015–December 2025

Item	Count
Retained newspaper article records	21,303,573
AERI country series	54
AERI country-month records	7,127
Records with at least 100 articles	6,488
Countries with matched AIRI and monthly CPI	53
Matched AIRI–CPI observations	6,912
Countries with lagged CPI, AIRI and observed FX changes	46
Three-variable dynamic-regression observations	5,830
Countries in two-lag Mean Group VAR	45
Matched recursive forecast origins, 2020–2024	2,612

Notes. Article and index counts come from the AERI comparison dataset; macroeconomic coverage is taken from *Selected Macro Indicators.xlsx*. Eritrea has no matching CPI series in the imported worksheet. Country-level VARs require at least 45 observations with complete lag information. The common forecast sample requires at least 48 complete training pairs per horizon and inflation outcomes through 2025.

The monthly regressions use CPI inflation, AIRI and exchange-rate depreciation. Countries enter an estimate only when the necessary observations are available. The forecast evaluation imposes additional requirements for the length of the estimation sample and complete outcomes at all four horizons. Country-name harmonisation and the resulting sample definitions are documented in the accompanying files.

4 Empirical Design

4.1 Sample formation and observed macroeconomic variables

The index and the macroeconomic series have different initial coverage. Newspaper archive counts describe the full 54-country production sample, while each empirical test uses observations for which its own regressors and outcomes are available. The first matching step links the index to year-on-year headline CPI inflation. This produces 53 countries and 6,912 country-month records over the study window. The regression including observed exchange-rate movements uses fewer countries and fewer months because its information requirements are stricter. The forecast exercise imposes additional minimum-history requirements to permit country-specific recursive estimation.

The monthly CPI and exchange-rate worksheets from *Selected Macro Indicators.xlsx* provide the macroeconomic observations. Inflation is recorded in percent year-on-year. Exchange-rate depreciation is the monthly change in the logarithm of units of domestic currency per US dollar, multiplied by one hundred. A positive sign denotes a depreciation. We calculate changes only for consecutive positive exchange-rate observations. The real-GDP workbook provides quarterly observed growth rather than a uniform observed monthly GDP series; we do not assign those quarterly values arbitrary monthly dates and include them in the monthly equations. The absence of a comparable complete M2 panel likewise determines the choice not to include money growth in the verified monthly specification.

The monthly empirical model uses the three variables that can be matched directly at the required frequency. Its coefficients describe that specified information set and should not be interpreted as estimates from a larger system with additional monetary and output controls. This definition also keeps the dynamic regressions and the forecast comparisons on a consistent empirical footing.

Country names are standardized before data matching. In the source archive, Cabo Verde and Cape Verde appear as separate raw labels. They refer to the same national economy and are combined for coverage purposes. We also reconcile alternative names for the Republic of the Congo and Democratic Republic of the Congo. Countries are counted

after harmonization. A report of 55 raw labels should therefore not be read as 55 distinct African economies.

4.2 Interpretation of the country-specific VAR

The panel-VAR literature contains several approaches to estimation and inference ([1, 12]). Tests of slope homogeneity such as Pesaran and Yamagata ([41]) address whether pooling restrictions are compatible with the data, while heterogeneous Granger-causality procedures allow cross-country differences in predictive precedence ([19]). These results inform the design, but the paper's primary reported coefficients come from the specified three-variable equations, not from an unreported causal-inference or structural identification procedure.

The country-specific system in equation (5) allows every lag coefficient to vary by country. For a two-lag, three-variable VAR, each equation contains six lagged regressors and a country-specific intercept. The coefficient on lagged AIRI in the inflation equation measures the linear association between the previous month's newspaper risk score and current year-on-year inflation, controlling for the previous two observations of all three variables. It does not have the interpretation of the effect of an exogenous newspaper intervention.

The report of the coefficient distribution is important for two reasons. First, averages can obscure a minority of countries with coefficients of the opposite sign. Second, the country-specific statistical precision is likely to vary with the number of observations, the variance of AIRI and inflation persistence. The arithmetic Mean Group average gives equal weight to each included country irrespective of its economic size or precision. It answers a different question from a pooled estimate that constrains the slope to be common.

The paper reports the 45-country distribution of first-lag AIRI coefficients and an average across those estimates. It does not infer a continent-wide structural impulse response from that average. The source results are sufficient to describe the sign and dispersion of reduced-form coefficients; structural identification would require additional assumptions about the joint innovations in inflation, exchange rates and news reporting.

4.3 Time effects, clustered inference and persistence

Year-on-year inflation contains persistent movements by construction, because adjacent observations share eleven months of the underlying price-level change. Economic shocks can also move annual inflation gradually over several months. The first lag of inflation is

therefore an essential control, but persistent outcomes can make standard errors sensitive to serial correlation. Clustering by country permits the residual process to be correlated within a national time series. Clustering by calendar month additionally permits a disturbance shared across countries in the same month. Both dimensions are relevant in a continent-wide study exposed to global food and fuel prices.

The original research documents strong common movement in a broader five-variable panel. In general, tests for cross-sectional dependence ([37]) and unit-root diagnostics in heterogeneous panels ([27, 39]) have different assumptions about common factors. We do not transfer numerical statistics from that earlier five-variable system to the re-estimated monthly sample; the inference reported here is based on the models and observations described in the preceding subsections.

Calendar-month fixed effects are particularly important around the global inflation episode in 2022–2023. A common price shock can generate simultaneous inflation articles and increases in CPI even when the country-specific narrative has no independent influence. By absorbing a separate mean outcome for each calendar month, the regression compares countries relative to the contemporaneous panel-wide movement. The AIRI coefficient consequently refers to variation remaining within a month and across countries after controlling for country intercepts and lagged covariates. It is not derived solely from the fact that both series rose during the same global episode.

There are limits to that control. Common shocks can have different intensities across countries, and a common factor with heterogeneous loadings is not identical to a calendar-month fixed effect. The large heterogeneity in African exchange-rate and monetary arrangements gives reason to expect such differences. The model should be interpreted as a dependence-aware conditional regression rather than a complete structural model of inflation formation. The full specifications are stated so that the interpretation of each coefficient can be traced to the variables included.

4.4 Regime equations and inference

The regime analysis addresses two outcomes and should not combine their coefficient estimates. In equation (7), AIRI itself is the dependent variable and the threshold coefficient measures a difference in the level of newspaper-reported inflation risk between states, conditional on covariates. For the inflation association, we estimate a separate interaction specification:

$$\pi_{ct} = \alpha_c + \lambda_t + \rho_1 \pi_{c,t-1} + \rho_2 \pi_{c,t-2} + \gamma d_{c,t-1} + \beta_L a_{c,t-1} + \Delta\beta\, a_{c,t-1} D_{c,t-1} + \eta D_{c,t-1} + v_{ct}, \quad (3)$$

where $D_{c,t-1} = \mathbf{1}\{\pi_{c,t-1} \geq 10\}$, β_L is the slope below 10 percent and $\beta_L + \Delta\beta$ is the slope in the double-digit regime. The inferential question is whether $\Delta\beta$ differs from zero under the stated standard-error estimator. A statistically significant coefficient within the high-inflation group would not, by itself, demonstrate that the two coefficients differ.

The analysis includes other cutoff values as sensitivity checks. The distinction between exploratory threshold comparisons and an independently estimated attention threshold is maintained. Because several cutoffs are assessed, we report the family-adjusted values where available. The 10 percent coefficient difference is significant at the nominal 10 percent level, but it is not significant at 5 percent; the threshold-family adjustment is also not significant at conventional levels. These are different statements and are given their proper statistical interpretation.

4.5 Forecast-origin information sets and performance measures

The forecast comparison follows the longstanding view that a simple inflation autoregression is an important benchmark for more elaborate models ([48, 6]). Under nested models, conventional equal-predictive-accuracy procedures require care because estimated extra parameters introduce noise even when the added regressor has no true forecast value ([15, 51]). We therefore distinguish regression evidence, adjusted nested-model evidence and actual reductions in realized forecast losses rather than collapsing them into one statement of superiority.

Our direct forecasts predict the value of year-on-year inflation h months after the forecast origin. At each country and horizon, three linear models are compared: a forecast from inflation history, a forecast including exchange-rate information, and a third forecast also including AIRI. Coefficients are re-estimated recursively as the training period expands. For a target at origin t and horizon h, a training pair dated s can be used only if its target observation $s + h$ is dated no later than t. This prevents the common error of training an h-step model on outcomes that had not occurred at the forecast origin.

Because countries and horizons have different missing-data patterns, the raw number of forecasts could vary across models. We address that by retaining the same country–origin observations for all four horizons and all competing specifications. Thus the headline comparison is not driven by an easier sample for one horizon or a different set of countries for the augmented model. The common evaluation set has 2,612 country–origin observations in 45 countries, with forecast origins from January 2020 through December 2024 and realized targets observed through December 2025.

Two summaries of predictive accuracy are needed. Pooled RMSE weights a forecast with a large squared error more heavily, irrespective of country, so countries with extreme

inflation can dominate the result. Equal-country summaries calculate each country's relative improvement first and then average across national series. The statistics need not agree. The difference is a substantive feature of the research question: is the aim to reduce total squared error in the pooled sample, or to describe performance in a typical included economy?

For each country c and horizon h, define

$$G_{ch} = 100\left(1 - \frac{\mathrm{RMSE}_{c,h}^{\mathrm{AR+FX+AIRI}}}{\mathrm{RMSE}_{c,h}^{\mathrm{AR+FX}}}\right). \tag{4}$$

A positive value indicates lower RMSE with AIRI. The pooled analogue replaces both country-specific RMSEs with pooled sample RMSEs. The paper also reports the number of countries for which $G_{ch} > 0$. We evaluate raw squared-loss differences with country-level HAC procedures and block-bootstrap calculations that retain common calendar movements; a favorable numerical difference should not be described as a reliably superior forecast when those tests do not support it.

The forecasts are historical retrospective comparisons using dated data files, not a real-time reconstruction of every release date and data revision. This qualification is confined to the description of the design and does not alter the interpretation of the observed forecast errors. In particular, none of the results is presented as evidence that an operational nowcasting system would have achieved the same accuracy before each official publication.

4.6 Country-specific dynamics

Let $y_{ct} = (\pi_{ct}, a_{ct}, d_{ct})'$, where π is year-on-year CPI inflation, a is AIRI and d is monthly exchange-rate depreciation. The country-specific reduced-form VAR is

$$y_{ct} = \boldsymbol{\alpha}_c + B_{1c}y_{c,t-1} + B_{2c}y_{c,t-2} + u_{ct}. \tag{5}$$

For each country with at least 45 usable observations, we estimate the inflation equation of the two-lag system and collect the coefficient on $a_{c,t-1}$. We report its mean, median and distribution across countries rather than impose one set of dynamic coefficients on the entire panel. This follows the heterogeneous-panel approach of Pesaran and Smith ([40]). Because country-specific estimation alone does not account for shocks shared by several economies, we use a separate specification with calendar-month effects to estimate the average conditional association.

4.7 A specification with common-month controls

The principal pooled diagnostic is

$$\pi_{ct} = \alpha_c + \lambda_t + \rho\pi_{c,t-1} + \beta a_{c,t-1} + \delta d_{c,t-1} + \varepsilon_{ct}, \tag{6}$$

The intercept α_c is specific to each country, while λ_t is a separate effect for each calendar month. These month effects absorb shocks common to the sample in a given month, including the 2022–2023 global inflation episode. We cluster standard errors by country and month. This allows errors to be correlated within countries over time and among countries in the same month. More general common-factor approaches are available for heterogeneous panels ([38]), but equation (6) provides a direct estimate of the association between lagged AIRI and inflation after controlling for the stated covariates and calendar shocks. The equation does not identify a causal effect of newspaper reporting.

We repeat the inflation–AIRI regression without exchange rates in the larger 53-country sample, and examine the influence of South Sudan and low-volume news months. For the regime analysis, we use the preceding month's CPI inflation to classify observations as below 10 percent or at least 10 percent. One equation compares the level of AIRI across the two states; a second allows its coefficient in the inflation equation to vary by state. We also compare later outcomes across country groups classified by their average inflation during 2015–2019.

4.8 Forecast comparison

We estimate direct country-specific forecasts of inflation one, three, six and twelve months ahead. The benchmark uses inflation history; the next model adds exchange-rate depreciation; the third adds AIRI as well. For each forecast origin between January 2020 and December 2024, the estimation window expands using only training pairs whose target month is no later than that origin. We require at least 48 completed training pairs for each country and horizon. A country–origin observation enters the common evaluation sample only when all four horizon forecasts can be produced. The competing models are fitted to the same training observations.[2]

We compare root mean squared errors (RMSE) over the pooled sample and within each country, as well as the number of countries with lower error after AIRI is added. We use a twelve-month calendar-block bootstrap to preserve common shocks across countries and serial dependence across forecast origins. For country-specific tests of raw squared-loss

[2]These retrospective exercises use the dated observations in the historical workbook. They are not reconstructions from archived publication-time data vintages.

differences, we apply Newey–West standard errors and adjust for multiple comparisons with the Holm procedure. We also compare forecast losses across inflation regimes defined at each forecast origin, using two-way-clustered standard errors. These comparisons concern realised forecast errors; they are different from the Clark–West adjustment for nested models ([15]).

5 Empirical Results

5.1 Distribution of newspaper coverage and inflation-related risk

The breadth of the newspaper archive is an advantage for the construction of a comparative index, but the distribution of coverage matters for interpretation. The full-period article count combines country-months with very different publication volumes. Some archives contain thousands of articles in a typical month, while other countries have short or intermittent runs of coverage. Article-weighted continent-wide averages would consequently be dominated by countries with large source collections. For that reason, the primary empirical units are country-month observations and the country-specific results are displayed separately from the pooled regression.

The denominator of AIRI includes every retained article, not merely those judged economically relevant. Changes in the share of economic reporting thus matter. During elections, security incidents or sporting events, a rise in other news categories can lower the risk score even when the number of inflation reports stays constant. Conversely, a deterioration in economic circumstances can increase both the count of inflation reports and the severity terms within them. These effects are not statistically identified as separate causes by the aggregate monthly score. The interpretation of the index follows from the construction: it measures the weighted prominence of reported inflation risk in the retained corpus.

The empirical sample has an additional selection effect. A country may be covered by the newspaper index but absent from a regression requiring an observed exchange-rate history. The reduced regression sample should not be described as a smaller newspaper corpus or taken to imply that the excluded economies lack inflation-risk reporting. Table 1 distinguishes the source coverage, CPI match, dynamic equation and recursive forecasting sample. Throughout the figures, the sample is identified explicitly rather than treated as a single fixed number of African economies.

5.2 Contemporaneous associations and timing

The country-level correlations provide an initial picture of whether the index and inflation tend to move together. Positive correlations in 40 of the 51 nonconstant country series indicate that co-movement is common, although its strength differs substantially. The average correlation of about 0.268 is not a forecast result and should not be equated with the coefficient from a lagged regression. In the pooled data, between-country variation and changes over time are mixed together. Country-by-country correlations remove some of the between-country difference, but are still sensitive to persistence, common shocks and the relative lengths of national series.

The annual medians provide a complementary view. Standardizing AIRI and CPI separately puts them in comparable plotted units without pretending that they measure the same magnitude. The global inflation episode around 2022 provides a useful reference period because both inflation and the attention devoted to prices changed markedly in many economies. The plot cannot by itself establish whether the news measure leads those changes. A lagged regression is needed to address temporal precedence, and even that remains a reduced-form association rather than evidence about how news affects economic decisions.

5.3 What the heterogeneous estimates reveal

The mean first-lag AIRI coefficient in the country-specific inflation equations is positive. More informative than the mean alone is the fact that 37 of 45 country coefficients have a positive sign, while eight are negative or zero. The cross-country distribution therefore does not resemble a single identical effect estimated with independent noise. There is genuine dispersion in the observed coefficients, reflecting both economic differences and the quality and composition of country-level reporting.

A positive country coefficient is not equivalent to a statistically significant one, and the cross-country average does not assign equal precision to each national regression. Figure 3 illustrates this dispersion. The pooled conditional regression in Table 2, with country and month controls, provides a separate dependence-aware summary: the estimate of 0.0355 has a two-way-clustered standard error of 0.0104. Under that specification, a one-point difference in lagged AIRI is associated with approximately 0.0355 percentage points of year-on-year inflation, holding the stated controls fixed. Its economic magnitude must be interpreted against the variation in AIRI rather than against the absolute number of newspaper articles.

The coefficient on the first inflation lag is close to one. Persistence is a defining feature of the year-on-year inflation series; it also makes marginal forecast improvements difficult

to obtain. An additional predictor may show a detectable regression relationship when averaged across many country-months and still have limited effect on the forecast errors of models that already use recent inflation. The results in the forecasting section are consistent with this distinction.

5.4 The broader panel and source-coverage sensitivity

The wider CPI sample is useful for examining dependence on the country selection, although it is not an alternative estimate of the identical three-variable model. Without the exchange-rate control, the AIRI coefficient is positive and marginally significant in the full 53-country sample. Its estimate changes when South Sudan is removed and when months with fewer than one hundred retained articles are excluded. These comparisons show that the inference depends in part on coverage and economic extremes. They should not be presented as proof that the monthly FX-controlled estimate is robust to every change in sample and specification.

The observed inflation distribution in the panel is highly uneven. South Sudan is an extreme case in the original results, and in a pooled squared-error criterion a relatively small number of observations can account for a large share of total loss. This makes it important not to select a preferred specification solely by whether excluding that country improves a particular forecasting statistic. We report the complete-sample estimates as the baseline and use country-equal evaluations to complement aggregate forecasts.

5.5 Inflation regimes and changes in measured newspaper risk

Two results from the double-digit analysis should be kept separate. The first concerns the level of the newspaper-risk index. After controlling for preceding CPI inflation, news volume and fixed effects, AIRI is 1.090 points higher following months in which annual inflation is at least 10 percent. This relationship remains positive under the reported controls for overall economic news composition. It is evidence that the risk-weighted content of the archive differs in the double-digit inflation environment. Whether that difference comes primarily from more inflation stories, stronger severity language or changes in the underlying newspapers cannot be resolved from the aggregate series alone.

The second result concerns the inflation regression. The lagged AIRI coefficient is estimated at 0.0172 below 10 percent inflation and at 0.0539 when the previous month's inflation is at least 10 percent. The estimated difference, 0.0368, has a standard error of 0.0195 and a two-sided p-value of 0.0589. It is significant at the nominal 10 percent level but not at 5 percent. That sentence is the appropriate statistical description. A separate

test of the individual high-inflation coefficient is not a substitute for the difference test.

The reporting result is not a measure of household expectations. There are circumstances in which newspapers may cover inflation more heavily because readers seek information; there are also circumstances in which food prices, currency changes and policy announcements simply generate more reportable economic events. With no independent attention series, the regression cannot distinguish those explanations. The term "state-dependent newspaper risk reporting" accurately characterizes what the evidence measures.

5.6 Why the forecast findings differ from the regressions

The three- and six-month pooled comparisons against AR+FX show small improvements when AIRI is added. At one and twelve months, the augmented specification is slightly less accurate. Pooled RMS errors depend strongly on country composition: the six-month advantage is materially reduced when Ghana is removed. In contrast, only about one third of the included countries improve at any horizon, and the mean of the country-specific gains is negative. This is a material finding for the interpretation of the continental result, not a contradiction that should be hidden.

Forecast losses are also subject to sampling variation. At four horizons in many countries, a handful of low p-values is expected even when no model systematically outperforms its benchmark. The corrected tests in the replication results do not identify any positive country–horizon gain that survives adjustment for the family of comparisons. The result is therefore best described as heterogeneous realized forecasting performance, with some substantial country-level point estimates but no general superiority claim.

The dependence of apparent forecast improvement on inflation conditions is similarly limited. High-inflation origins display a favorable six-month pooled gain, but the formal difference in normalized squared-loss performance is not statistically significant. The regression interaction result and the regime-forecast comparison answer different questions and use different loss functions. Neither should be interpreted as definitive evidence that changing attention causes forecast accuracy to improve when inflation crosses 10 percent.

5.7 Implications for economic surveillance

The practical question is what an institution obtains from AIRI that it would not obtain merely by watching current CPI inflation. The index is a documented record of how domestic newspapers discuss inflation-related threats. It provides a regularly updated, comparable form of evidence about the prominence and severity of reported risks under

a common scoring system. The regressions show that this record is associated with subsequent inflation after accounting for some observable macroeconomic information. They do not establish that it is a replacement for published CPI statistics, a causal mechanism of inflation, or a generally more accurate point-forecast model.

The country differences imply that AIRI should be interpreted in light of source coverage and the national inflation environment. For countries with persistent zero scores, little information can be extracted from changes in the series. For countries where inflation articles are frequent and the score varies meaningfully over time, the index offers a richer record of local risk narratives. These are measurement properties; whether they translate into accurate inflation predictions remains an empirical question addressed by the reported forecast exercises.

5.8 Cross-country patterns

Of the 51 countries with at least 36 matched, non-constant observations, 40 have a positive correlation between AIRI and CPI inflation. The mean country correlation is 0.268 and the median is 0.270. Figure 1 shows the variation across countries. These correlations describe contemporaneous co-movement. They do not reveal whether news coverage precedes price changes or reacts to inflation already observed.

Figure 2 plots the annual medians of AIRI and CPI inflation after standardising each series separately. Standardisation permits a comparison of their timing without placing the AIRI score and the CPI inflation rate on the same numerical scale.

5.9 Dynamic estimates and robustness

Table 2 reports the estimates of equation (6) for 5,808 observations in the 45-country forecast sample. The coefficient on lagged AIRI is 0.0355 (two-way clustered standard error 0.0104; $p = 0.0006$). CPI inflation is highly persistent: its first-lag coefficient is 0.9612. The estimated coefficient on lagged exchange-rate depreciation is 0.0375 (standard error 0.0129). These estimates describe conditional month-to-month relationships. The AIRI coefficient is small relative to the variation in CPI inflation across countries.

In the country-specific three-variable VAR(2) estimates, the mean coefficient on the first lag of AIRI is 0.0382 (cross-country t-test $p = 0.0041$), and the median is 0.0276. The coefficient is positive in 37 of 45 countries (Figure 3). These estimates allow dynamic relationships to differ by country; the cross-country t-test is a descriptive summary and does not adjust for shocks common to several countries. We base inference about the common conditional AIRI coefficient on the month-effects regression in Table 2.

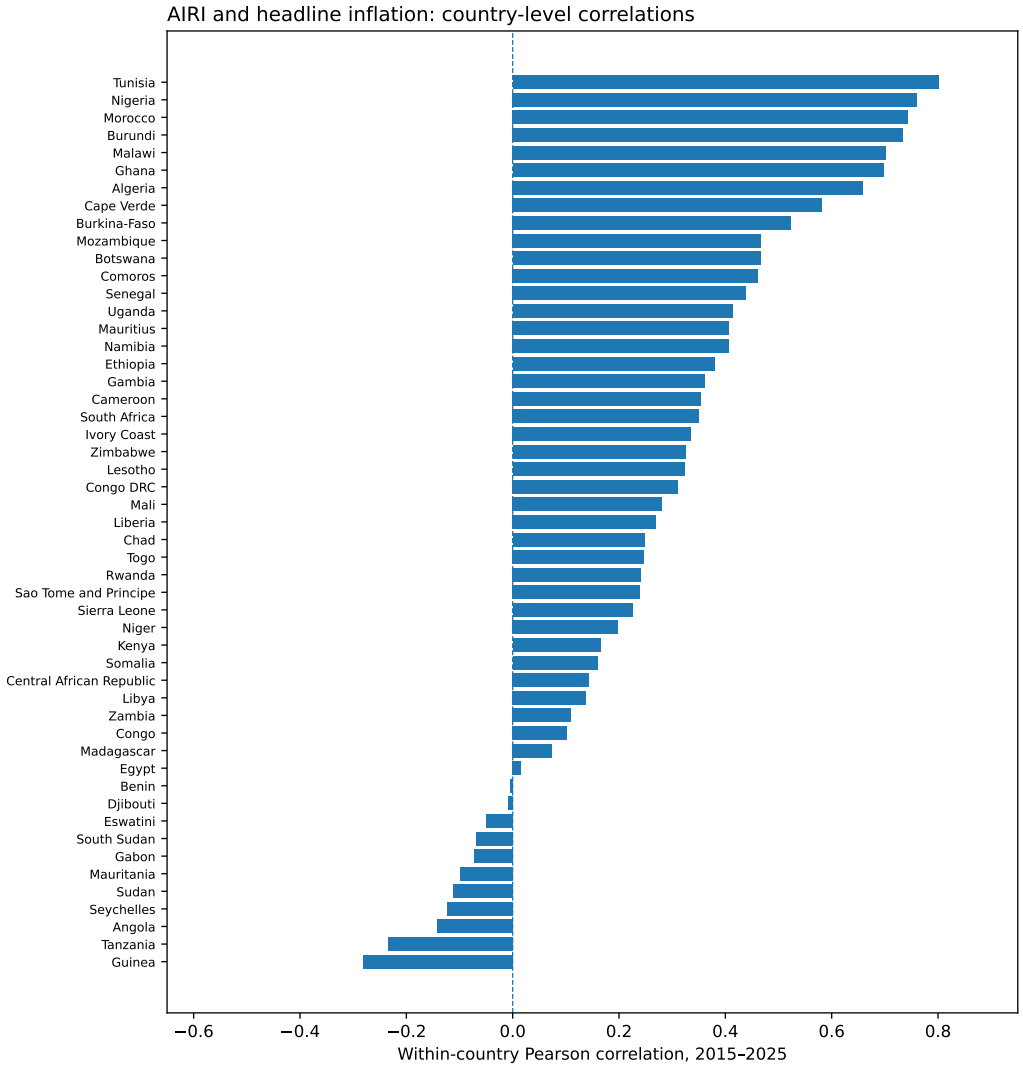


Figure 1: Country-level correlations between AIRI and observed CPI inflation

Notes. Pearson correlations use matched monthly AIRI and CPI observations from 2015–2025. Countries with fewer than 36 observations or constant series are excluded. The correlations are descriptive; standard errors are not adjusted for serial dependence.

Table 2: Monthly inflation regressions: country and calendar-month effects

Regressor	Coefficient	Two-way SE	Two-sided p
Inflation, lag 1	0.9612	0.0050	< 0.001
AIRI, lag 1	0.0355	0.0104	0.0006
Exchange-rate depreciation, lag 1	0.0375	0.0129	0.0038
Countries			45
Country-month observations			5,808
Country and calendar-month effects			Included

Notes. The dependent variable is year-on-year headline CPI inflation, in percentage points. Depreciation is $100\Delta \log e$, with the exchange rate measured in local currency per US dollar. Standard errors are clustered by country and calendar month. Estimates use the matched country-month AIRI and macroeconomic data.

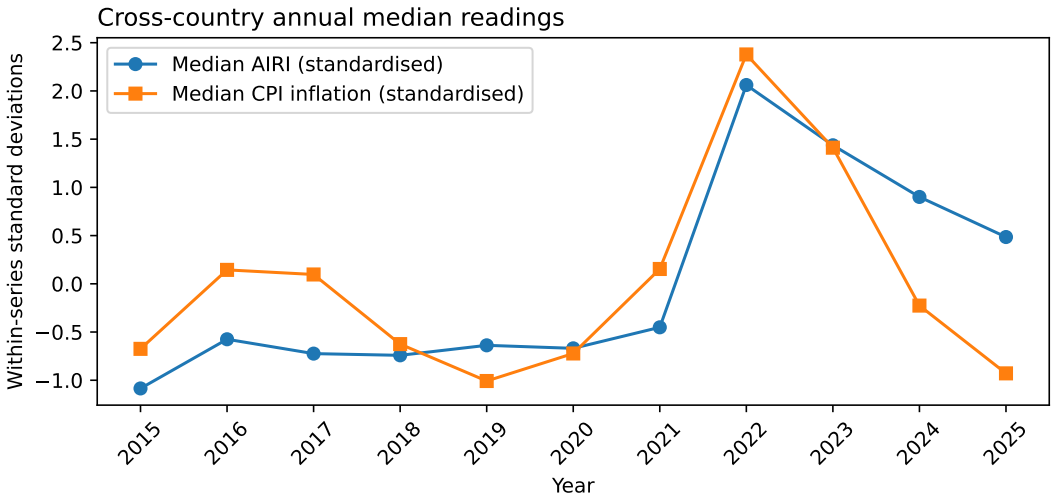


Figure 2: Standardised annual medians of AIRI and CPI inflation

Notes. Each annual series is formed from available country-month observations and standardised using its own 2015–2025 mean and standard deviation. The comparison concerns the timing of the two measures, not their levels or units.

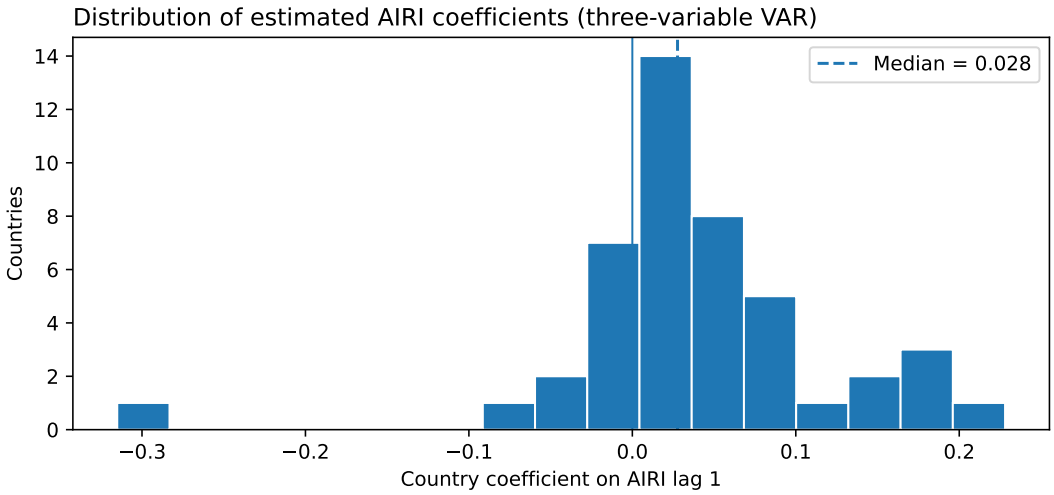


Figure 3: Country-level AIRI coefficients in three-variable VAR(2) inflation equations

Notes. Each inflation equation contains two lags of CPI inflation, AIRI and observed exchange-rate depreciation, plus an intercept. The plot covers the 45 countries with sufficient observations. The coefficients are reduced-form associations, not structural effects.

In the larger 53-country sample, a regression of inflation on its lag, lagged AIRI and country and calendar-month effects gives an AIRI coefficient of 0.2621 (two-way clustered standard error 0.1368; $p = 0.0554$). The coefficient falls to 0.2049 when South Sudan is excluded ($p = 0.1199$), and rises to 0.3180 when months with fewer than 100 articles are excluded ($p = 0.0515$). Unlike Table 2, these regressions do not include exchange-rate movements and are estimated on different observations. The estimates are therefore informative about sample sensitivity, not directly comparable coefficient changes within a single specification.

For a second comparison, we classify countries by their average inflation during 2015–2019. The median of these averages is 4.61 percent. In the 45-country panel for 2020–2025, the estimated interaction between AIRI and membership in the historically higher-inflation group is 0.0298 (two-way clustered standard error 0.0254; $p = 0.241$). The next section reports the separate within-country comparison based on whether inflation in the preceding month reached 10 percent.

5.10 Double-digit inflation and inflation-related newspaper risk

We divide the monthly observations according to whether CPI inflation in the preceding month was below 10 percent or had reached double digits. The classification uses $\pi_{c,t-1}$, while the outcome is the current newspaper-risk score, AIRI_{ct}. We estimate

$$\text{AIRI}_{ct} = \alpha_c + \lambda_t + \rho\pi_{c,t-1} + \delta\mathbf{1}\{\pi_{c,t-1} \geq 10\} + \phi \log N_{ct} + e_{ct}, \tag{7}$$

In this equation, N_{ct} is the number of retained articles, and α_c and λ_t are country and calendar-month effects. Standard errors are clustered by country and month. The coefficient δ is the difference in AIRI between the two inflation states after controlling for the preceding inflation rate and the size of the newspaper archive. It measures association, not an effect of inflation on editorial decisions.

In the matched 46-country sample, $\hat{\delta} = 1.090$ AIRI points (standard error 0.394; $p = 0.0056$). Adding the share of economically relevant articles and their average severity reduces the estimate slightly to 1.040 ($p = 0.0018$). Those additional variables cover all economic news, not just inflation-related stories. Accordingly, the results concern the weighted inflation-risk score as a whole; they do not separately estimate the effects of inflation-news frequency and inflation-specific severity.

The positive difference persists when the 2022–2023 observations are excluded and when countries with constant AIRI series are removed. It is larger in 2015–2019 than in 2020–2025. The 10 percent cutoff denotes double-digit inflation and was not estimated

Table 3: AIRI and the double-digit inflation environment

Specification	Difference	SE	p-value
Country and calendar-month effects	1.206	0.401	0.0026
Add lagged CPI inflation	1.099	0.401	0.0061
Add current log article volume	1.090	0.394	0.0056
Also control economic-news share and severity	1.040	0.332	0.0018

Notes. The outcome is AIRI in index points. Double-digit inflation means CPI inflation in month $t-1$ is at least 10 percent. Standard errors are clustered by country and calendar month; all specifications include country and calendar-month effects. The sample contains 5,779 observations from 46 countries. The economic-news controls cover all economically relevant articles, not the inflation subset.

from a model of household attention. Supplementary results use cutoffs of 4, 8, 12 and 15 percent. Across the five cutoffs considered, the Holm-adjusted p-value for the 10 percent news-reporting result is 0.0169.

We next allow the lagged AIRI coefficient in the inflation equation to differ across the same two inflation states. Controlling for two CPI lags, lagged exchange-rate depreciation and country and calendar-month effects gives AIRI coefficients of 0.0172 below 10 percent and 0.0539 at or above 10 percent. Their difference is 0.0368 (standard error 0.0195; two-sided $p = 0.0589$), significant at the 10 percent level but not at 5 percent. The Supplementary Information reports the estimates for other inflation cutoffs. The comparison does not measure household attention directly.

Table 4: Lagged AIRI and inflation across the 10 percent cutoff

Inflation state	AIRI coefficient	Two-way SE
Below 10 percent	0.0172	0.0085
At least 10 percent	0.0539	0.0188
Difference	0.0368	0.0195

Notes. Dependent variable: year-on-year CPI inflation. Inflation states are defined by CPI inflation in the previous month. All regressions include two CPI lags, lagged observed exchange-rate depreciation and country and calendar-month effects (5,779 observations, 46 countries). The coefficient difference has a two-sided p-value of 0.0589, significant at 10 percent but not at 5 percent. Other cutoffs and multiple-testing adjustments appear in the Supplementary Information.

5.11 Direct forecasts at four horizons

We forecast year-on-year CPI inflation directly at $h = 1, 3, 6$, and 12 months for each country. The first model uses inflation at the forecast origin, the second adds exchange-rate depreciation, and the third adds AIRI. Each equation is re-estimated as new observations enter the sample. Training observations are included only if their outcome has occurred by the forecast origin; at least 48 completed pairs are required for each horizon. We evaluate

the models on 2,612 common country–origin observations from 45 countries. Forecast origins run from January 2020 to December 2024, and the latest evaluated inflation outcome is December 2025.

Table 5: Common-sample out-of-sample forecast accuracy at four horizons

h	n	AR	AR+FX	AR+FX+AIRI	Gain (%)	Countries
1	2,612	1.406	1.424	1.428	-0.26	15/45
3	2,612	3.087	3.153	3.095	+1.85	15/45
6	2,612	5.238	5.267	5.083	+3.49	15/45
12	2,612	7.952	8.018	8.037	-0.24	14/45

Notes. Gain equals $100(1 - \text{RMSE}_{AR+FX+AIRI}/\text{RMSE}_{AR+FX})$; positive values indicate lower RMSE after AIRI is added. The final column counts countries with a reduction in country-specific RMSE. Each model uses the same country–origin sample. The historical workbook does not provide archived publication-time data vintages.

The AIRI-augmented model has a lower pooled RMSE than the AR+FX benchmark at three and six months, but a higher RMSE at one and twelve months. Its six-month pooled gain of 3.49 percent changes to −0.06 percent when Ghana is removed. At that horizon only 15 of the 45 countries record lower RMSE after AIRI is added. Average country-specific gains are negative at every horizon, from −1.12 percent at one month to −7.84 percent at twelve months. The calendar-block bootstrap confidence intervals for these equal-country averages are also negative. After Holm adjustment, no positive country–horizon loss difference is statistically significant.

5.12 Forecast accuracy across inflation regimes

For each forecast origin we use the CPI inflation rate recorded at that date to distinguish observations below 10 percent from those at or above 10 percent. Thus future inflation outcomes do not enter the regime definition. We also repeat the comparison using the mean inflation rate over the previous twelve months and a classification of countries based on their 2015–2019 inflation histories.

At six months the pooled RMSE gain from AIRI is 5.99 percent in double-digit inflation observations and −2.64 percent in lower-inflation observations. The formal test of the difference in normalized squared losses gives $p = 0.299$, and the alternative regime definitions likewise show no statistically significant difference. In the matched monthly inflation regression, however, the coefficient on lagged AIRI remains positive at 0.0355 (two-way standard error 0.0104; $p = 0.0006$). The regression and forecasting results concern different aspects of the data: AIRI has a positive conditional association with subsequent inflation, but does not consistently lower forecast error.

Table 6: Conditional forecast accuracy for high- and low-inflation origins

h	$n(< 10\%)$	Gain low (%)	$n(\geq 10\%)$	Gain high (%)	Gap p
1	1982	-3.24	630	+2.46	0.286
3	1982	-1.32	630	+3.53	0.752
6	1982	-2.64	630	+5.99	0.299
12	1982	-3.18	630	+1.07	0.131

Notes. Gains compare AR+FX with AR+FX+AIRI using pooled RMSE within each state. The last column reports the two-way-clustered test of the difference in normalized squared-loss differentials, by country and forecast-origin month. The 10 percent cutoff classifies inflation at the origin; it was not preregistered.

6 Discussion and Conclusion

6.1 Interpretation and limits of the evidence

The finding that lagged AIRI is positively associated with subsequent inflation is economically interpretable but admits more than one explanation. A newspaper can report price developments that are already visible to consumers but have not yet entered the published CPI. It can report anticipated policy changes that will affect prices later. It can also describe general economic difficulties that correlate with omitted inflation pressures. The monthly regressions cannot discriminate fully among these channels. Their contribution is to establish the association while conditioning on inflation history, observed exchange-rate movements, country effects and shocks common to a given month.

The strength of the paper is the separation of measurement and evaluation. AIRI is defined before it is used to explain CPI, and its weights are not fitted to produce a desired regression coefficient. The resulting estimate therefore has a clearer interpretation than an index whose dictionary or weights were selected to maximize in-sample forecasting fit. At the same time, prespecification alone does not establish measurement validity. Newspaper selection, dates, language rules and classification thresholds remain part of the empirical object. Source-based sensitivity tests and independent article classification would be useful dimensions of further validation, but they are not represented here as completed results.

The evidence on double-digit inflation sharpens the descriptive finding. AIRI itself is higher following periods of double-digit inflation. Its lagged coefficient in the inflation equation is also numerically larger in that regime, with a nominal p-value of 0.0589 for the difference. These results document regime-associated changes in reporting and predictive association. They do not justify identifying a household-attention threshold at 10 percent, and the estimates do not imply a discontinuous change in the structural inflation process at precisely that value.

The forecasting exercise shows that information in a regressor and improvements in a forecast are not equivalent. With a highly persistent target such as year-on-year inflation, small regression relationships can be detectable with many observations yet have little effect on squared forecast errors. AIRI can be statistically informative under the reduced-form specification without being reliably superior to the autoregressive forecast in the average included country. Both results are part of the empirical record and should inform the interpretation of the index.

6.2 Relation to existing measurement studies

The paper belongs to the class of text-based economic measurement studies in which construction and validation are closely linked. Unlike a general economic uncertainty index, AIRI restricts the numerator to inflation-related articles and incorporates an ordered risk-severity score. Unlike a direct measure of consumer inflation expectations, it uses editorially published text and cannot reveal how individuals change their beliefs. And unlike purely predictive text models, its aggregation can be written as an explicit function of article categories and predefined weights. These distinctions define what the index measures and the kinds of validation that are relevant.

Cross-country coverage is substantively important. A national index may be tailored to its country's institutions, vocabulary and newspaper market. A common rule used in a large African sample creates a basis for comparing the time pattern of inflation-related reporting, while also exposing country-level differences in data depth. The comparisons cannot be made reliable by normalization alone; they require clear sample definitions, stable source treatment and careful interpretation of observations with little coverage. The empirical sections document how coverage and estimation samples change with the variables and tests used.

6.3 Conclusion

This study constructs AIRI from newspaper reports of inflation-related risks across African economies and relates the resulting country-month series to observed CPI inflation. The score combines relevance screening, an inflation-specific classification, ordered severity and textual prominence, divided by total retained article volume. The archive covers 54 country series during 2015–2025, while the monthly macroeconomic regressions use the countries and months for which the required observations are available.

Lagged AIRI has a positive association with inflation in the matched monthly specification after controlling for inflation persistence, exchange-rate depreciation and country

and calendar-month effects. Country-specific estimates vary considerably. The news-risk score is higher following double-digit inflation, and the AIRI coefficient in the inflation equation is larger in that environment, with the difference significant at 10 percent but not at 5 percent. These findings concern newspaper risk reporting and reduced-form inflation dynamics, not a causal effect of newspaper coverage on prices or a measured behavioral attention mechanism.

Forecast comparisons add a useful qualification. Incorporating AIRI lowers RMSE for some countries and at some horizons, but it does not consistently improve forecasts over conventional inflation and exchange-rate benchmarks. This does not invalidate the measurement contribution. It establishes a boundary on the index's use as a general point-forecasting predictor. AIRI is most defensibly interpreted as a systematic measure of inflation-related narratives, accompanied by evidence about its association with subsequent inflation and the substantial heterogeneity of that association across African economies.

Data and Replication

The replication package contains monthly AIRI scores and article totals, matched CPI and exchange-rate series, the country-name harmonisation rules, estimation code, country coefficients and forecast-origin predictions. The original newspaper texts and classification dictionaries are stored separately from this analytical release.

References

[1] Abrigo, M. R. M., and Love, I. (2016). Estimation of panel vector autoregression in Stata. *Stata Journal*, 16(3), 778–804.

[2] Ahir, H., Bloom, N., and Furceri, D. (2022). The World Uncertainty Index. *NBER Working Paper* 29763.

[3] Alper, C. E., Hobdari, N. A., and Uppal, A. (2016). Food inflation in Sub-Saharan Africa: Causes and policy implications. *IMF Working Paper* 16/247.

[4] Angelico, C., Marcucci, J., Miccoli, M., and Quarta, F. (2022). Can we measure inflation expectations using Twitter? *Journal of Econometrics*, 228(2), 259–277.

[5] Araujo, G. S., and Gaglianone, W. P. (2023). Machine learning methods for inflation forecasting in Brazil: New contenders versus classical models. *Latin American Journal of Central Banking*, 4(2), 100087.

[6] Atkeson, A., and Ohanian, L. E. (2001). Are Phillips curves useful for forecasting inflation? *Federal Reserve Bank of Minneapolis Quarterly Review*, 25(1), 2–11.

[7] Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.

[8] Barbaglia, L., Consoli, S., and Manzan, S. (2023). Forecasting with economic news. *Journal of Business & Economic Statistics*, 41(3), 708–719.

[9] Barnichon, R., and Peiris, S. J. (2007). Sources of inflation in Sub-Saharan Africa. *IMF Working Paper* WP/07/32.

[10] Blinder, A. S., Ehrmann, M., de Haan, J., and Jansen, D.-J. (2024). Central bank communication with the general public: Promise or false hope? *Journal of Economic Literature*, 62(2), 425–457.

[11] Bracha, A., and Tang, J. (2025). Inflation levels and (in)attention. *Review of Economic Studies*, 92(3), 1564–1594.

[12] Canova, F., and Ciccarelli, M. (2013). Panel vector autoregressive models: A survey. *Advances in Econometrics*, 32, 205–246.

[13] Cavallo, A., and Rigobon, R. (2016). The Billion Prices Project: Using online prices for measurement and research. *Journal of Economic Perspectives*, 30(2), 151–178.

[14] Choi, H., and Varian, H. (2012). Predicting the present with Google Trends. *Economic Record*, 88(s1), 2–9.

[15] Clark, T. E., and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.

[16] Coibion, O., and Gorodnichenko, Y. (2012). What can survey forecasts tell us about information rigidities? *Journal of Political Economy*, 120(1), 116–159.

[17] Coibion, O., and Gorodnichenko, Y. (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review*, 105(8), 2644–2678.

[18] Coibion, O., Georgarakos, D., Gorodnichenko, Y., and van Rooij, M. (2023). How does consumption respond to news about inflation? Field evidence from a randomized control trial. *American Economic Journal: Macroeconomics*, 15(3), 109–152.

[19] Dumitrescu, E.-I., and Hurlin, C. (2012). Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29(4), 1450–1460.

[20] Durevall, D., and Sjö, B. (2012). The dynamics of inflation in Ethiopia and Kenya. *African Development Bank Working Paper* 151.

[21] Elshendy, M., and Fronzetti Colladon, A. (2017). Big data analysis of economic news: Hints to forecast macroeconomic indicators. *International Journal of Engineering Business Management*, 9.

[22] Friedman, M. (1968). The role of monetary policy. *American Economic Review*, 58(1), 1–17.

[23] García, D. (2013). Sentiment during recessions. *Journal of Finance*, 68(3), 1267–1300.

[24] Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3), 535–574.

[25] Ghirelli, C., Pérez, J. J., and Urtasun, A. (2019). A new economic policy uncertainty index for Spain. *Economics Letters*, 182, 64–67.

[26] Goshima, K., Ishijima, H., Shintani, M., and Yamamoto, H. (2020). Forecasting Japanese inflation with a news-based leading indicator of economic activities. *Studies in Nonlinear Dynamics & Econometrics*. doi:10.1515/snde-2019-0117.

[27] Im, K. S., Pesaran, M. H., and Shin, Y. (2003). Testing for unit roots in heterogeneous panels. *Journal of Econometrics*, 115(1), 53–74.

[28] Jerven, M. (2013). *Poor Numbers: How We Are Misled by African Development Statistics and What to Do about It*. Cornell University Press.

[29] Korenok, O., Munro, D., and Chen, J. (2026). Inflation and attention thresholds. *Review of Economics and Statistics*, 108(3), 842–850. doi:10.1162/rest_a_01402.

[30] Larsen, V. H., Thorsrud, L. A., and Zhulanova, J. (2021). News-driven inflation expectations and information rigidities. *Journal of Monetary Economics*, 117, 507–520.

[31] Love, I., and Zicchino, L. (2006). Financial development and dynamic investment behavior: Evidence from panel VAR. *Quarterly Review of Economics and Finance*, 46(2), 190–210.

[32] Mankiw, N. G., and Reis, R. (2002). Sticky information versus sticky prices: A proposal to replace the New Keynesian Phillips curve. *Quarterly Journal of Economics*, 117(4), 1295–1328.

[33] Mishkin, F. S. (2007). Inflation dynamics. *International Finance*, 10(3), 317–334.

[34] Nguyen, A. D. M., Dridi, J., Unsal, D. F., and Williams, O. (2017). On the drivers of inflation in Sub-Saharan Africa. *International Economics*, 151, 71–84.

[35] Nyman, R., Kapadia, S., and Tuckett, D. (2021). News and narratives in financial systems: Exploiting big data for systemic risk assessment. *Journal of Economic Dynamics and Control*, 127, 104119.

[36] Orphanides, A. (2001). Monetary policy rules based on real-time data. *American Economic Review*, 91(4), 964–985.

[37] Pesaran, M. H. (2004). General diagnostic tests for cross section dependence in panels. *CESifo Working Paper* 1229.

[38] Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4), 967–1012.

[39] Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265–312.

[40] Pesaran, M. H., and Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, 68(1), 79–113.

[41] Pesaran, M. H., and Yamagata, T. (2008). Testing slope homogeneity in large panels. *Journal of Econometrics*, 142(1), 50–93.

[42] Phelps, E. S. (1968). Money-wage dynamics and labour-market equilibrium. *Journal of Political Economy*, 76(4), 678–711.

[43] Razafimahefa, I. F. (2012). Exchange rate pass-through in Sub-Saharan African economies and its determinants. *IMF Working Paper* 12/141.

[44] Salisu, A., Salisu, S., and Salisu, S. (2024). A news-based economic policy uncertainty index for Nigeria. *Quality & Quantity*, 58(5), 4987–5002.

[45] Shapiro, A. H., Sudhof, M., and Wilson, D. J. (2022). Measuring news sentiment. *Journal of Econometrics*, 228(2), 221–243.

[46] Shiller, R. J. (2017). Narrative economics. *American Economic Review*, 107(4), 967–1004.

[47] Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3), 665–690.

[48] Stock, J. H., and Watson, M. W. (1999). Forecasting inflation. *Journal of Monetary Economics*, 44(2), 293–335.

[49] Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.

[50] Weber, M., D'Acunto, F., Gorodnichenko, Y., and Coibion, O. (2022). The subjective inflation expectations of households and firms: Measurement, determinants, and implications. *Journal of Economic Perspectives*, 36(3), 157–184.

[51] West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica*, 64(5), 1067–1084.

A Source Coverage and Sample Accounting

The country table below reports distinct observed country-month records, retained newspaper article totals and the mean AIRI score over the historical period. A month with a small number of retained articles is counted rather than imputed; the column headed "low coverage" records months with fewer than 100 articles. The figures were calculated from the archived country-month aggregate file. They describe the media collection, not the number of countries with complete CPI or exchange-rate observations.

Table 7: Newspaper archive coverage by country, 2015–2025

Country	Months	Articles	Mean AIRI	Low coverage
Algeria	132	730,625	7.70	0
Angola	132	235,807	4.10	0
Benin	132	264,265	2.02	0
Botswana	132	118,124	7.75	0
Burkina-Faso	132	292,634	2.05	0
Burundi	132	48,270	6.61	0
Cabo Verde	132	92,443	2.17	1
Cameroon	132	320,390	2.63	0
Central African Republic	132	53,300	4.66	3
Chad	132	48,287	3.46	36
Comoros	132	39,882	5.85	2
Congo	132	115,038	2.89	0
Congo DRC	132	146,368	2.92	0
Djibouti	132	6,870	10.36	128
Egypt	132	168,849	4.92	0
Equatorial Guinea	132	15,936	0.00	59
Eritrea	132	25,535	9.04	0
Eswatini	132	30,294	4.85	1
Ethiopia	132	146,573	8.63	0
Gabon	132	130,689	3.62	0
Gambia	132	98,038	5.33	0

Table A1 continued

Country	Months	Articles	Mean AIRI	Low coverage
Ghana	132	1,526,851	6.13	0
Guinea	132	263,769	2.06	0
Guinea-Bissau	131	262	0.00	131
Ivory Coast	132	333,206	0.75	0
Kenya	132	354,881	3.43	0
Lesotho	132	47,349	6.40	0
Liberia	132	71,839	7.20	14
Libya	132	428,722	0.94	0
Madagascar	132	191,290	1.84	0
Malawi	132	278,827	8.77	0
Mali	132	676,319	3.39	0
Mauritania	132	32,345	0.08	0
Mauritius	132	171,856	2.06	0
Morocco	132	1,992,928	2.21	0
Mozambique	132	136,588	3.15	0
Namibia	132	143,093	5.41	0
Niger	132	111,343	1.36	0
Nigeria	132	6,043,761	8.69	0
Rwanda	132	42,822	2.34	22
Sao Tome and Principe	132	8,638	3.30	112
Senegal	132	985,533	1.10	0
Seychelles	132	3,890	19.52	130
Sierra Leone	132	137,376	9.15	0
Somalia	132	142,649	0.39	0
South Africa	132	873,015	4.55	0
South Sudan	132	72,985	6.48	0
Sudan	132	1,160,540	1.00	0
Tanzania	132	90,654	1.13	0
Togo	132	72,963	6.25	0
Tunisia	132	463,252	6.35	0
Uganda	132	578,704	4.28	0
Zambia	132	184,707	4.62	0
Zimbabwe	132	552,399	10.62	0

Notes. The underlying source has both “Cape Verde” and “Cabo Verde” labels; these are grouped as one country. Article totals are taken from the country-month production export and do not refer to unique

readers or households. The index is in scored risk units per 100 retained articles.

B Country-Specific Forecast Comparisons

Table 8 reports the reduction in RMSE when AIRI is added to the AR+FX specification, separately for each country and horizon. Positive entries favor AIRI. Forecast origins and evaluation cases match those in the main paper. Individual positive gains are descriptive; none survives the Holm adjustment across the country–horizon comparisons.

Table 8: Country forecast accuracy: AR+FX versus AR+FX+AIRI (gain in %)

Country	$h = 1$	$h = 3$	$h = 6$	$h = 12$

Notes. Gain $= 100(1 - \text{RMSE}_{AR+FX+AIRI}/\text{RMSE}_{AR+FX})$. Each country is tested at four horizons on the matched forecast origins. Blank values indicate insufficient observations. These are retrospective historical-file forecasts rather than archived publication-vintage forecasts.

C Inflation Regime Sensitivity

The primary 10 percent result is shown alongside alternative thresholds and selected subsamples. The difference column is the coefficient on the AIRI-by-regime interaction in an inflation equation with country and calendar-month effects and the lagged covariates specified in the main text. The associated p-values are two-sided. The adjustment for comparing thresholds is a distinct statistic from the unadjusted p-value.

Table 9: Threshold dependence of the AIRI coefficient

Threshold	$\widehat{\beta}_L$	$\widehat{\beta}_H$	Difference	p
4%	0.0213	0.0342	0.0129	0.2433
8%	0.0239	0.0409	0.0170	0.2955
10%	0.0172	0.0539	0.0368	0.0589
12%	0.0124	0.0661	0.0537	0.0247
15%	0.0190	0.0548	0.0358	0.1437

Notes. Regimes are based on inflation in the previous month. The primary 10% difference has $p = 0.0589$, significant at 10% but not at 5%. The 10% threshold-family Holm-adjusted p-value is approximately 0.236. These are regression association comparisons, not tests of household attention.

D Additional Figures

E Reading the Reproducibility Files

The analytical replication archive accompanying the manuscript includes the country-month AIRI and related risk-pillar data, extracted monthly CPI and exchange-rate observations, matched panels, code for the monthly analyses and the recursive forecast-origin predictions. The source archive and this appendix should be read together: Table A1 documents coverage, Table A2 shows country-specific forecasting effects, and the sensitivity table records the alternative thresholds examined. The upstream newspaper texts, source registry and exact dictionary matches are separate production-system inputs and are not reconstructed from these aggregate exports. Accordingly, article-level classification precision and recall are not claimed as outcomes of the statistical analyses reported here.

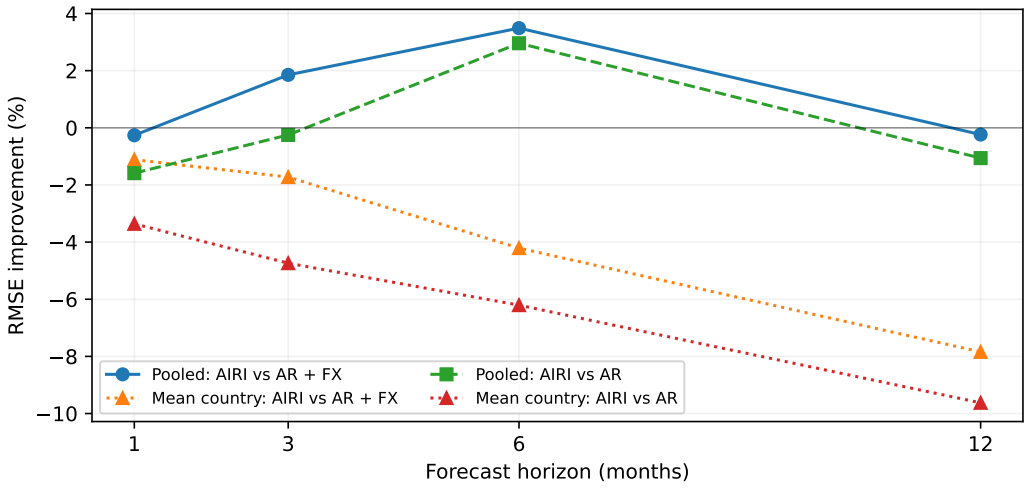


Figure 4: Aggregate and country-average improvements from AIRI at alternative forecast horizons

Notes. The same country-origin evaluation set is used at each horizon. The difference between pooled and country-average results reflects the weighting of extreme errors in pooled RMSE.

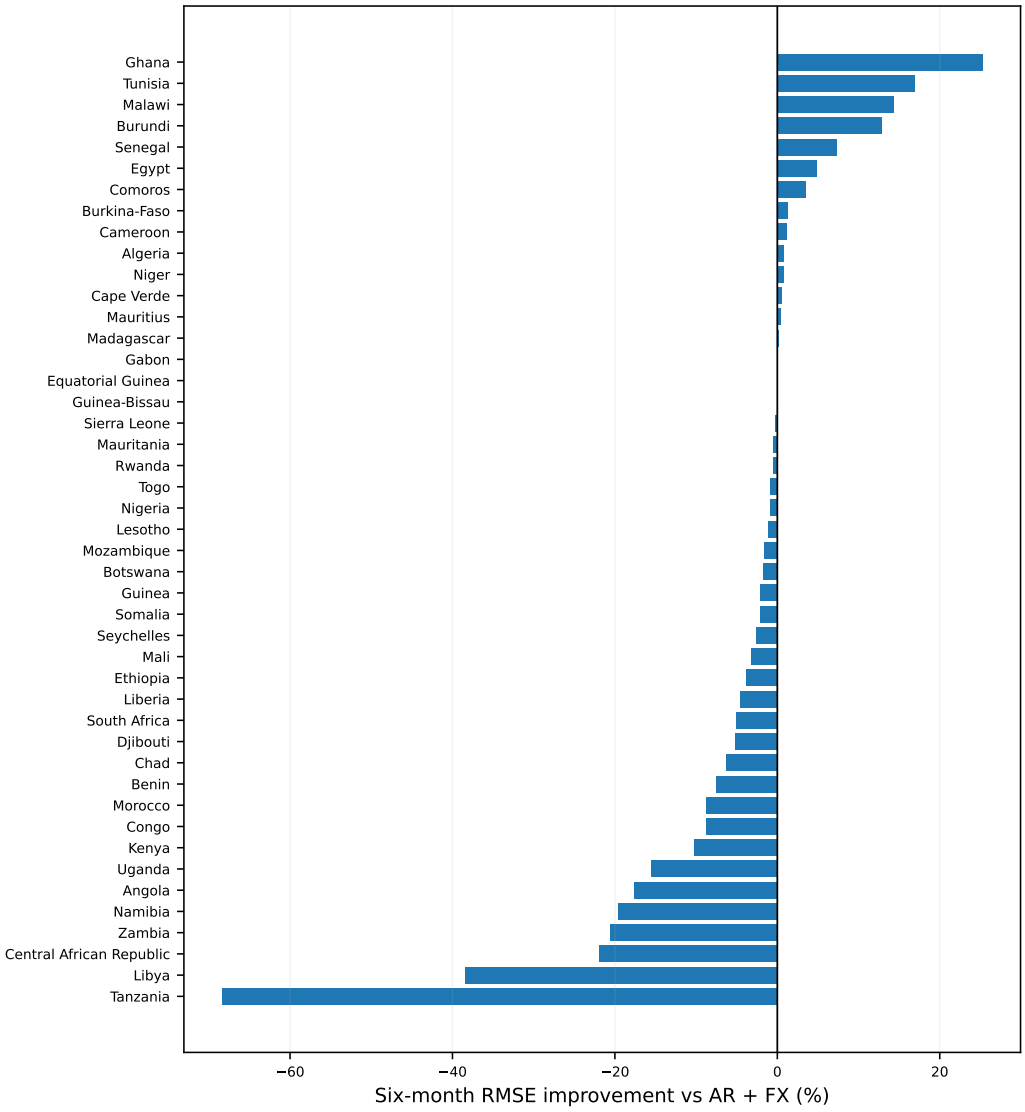


Figure 5: Country-specific six-month forecasting comparisons

Notes. The plotted values are retrospective differences in RMSE. Selected large improvements should not be equated with dependence-robust statistical superiority after multiplicity adjustment.

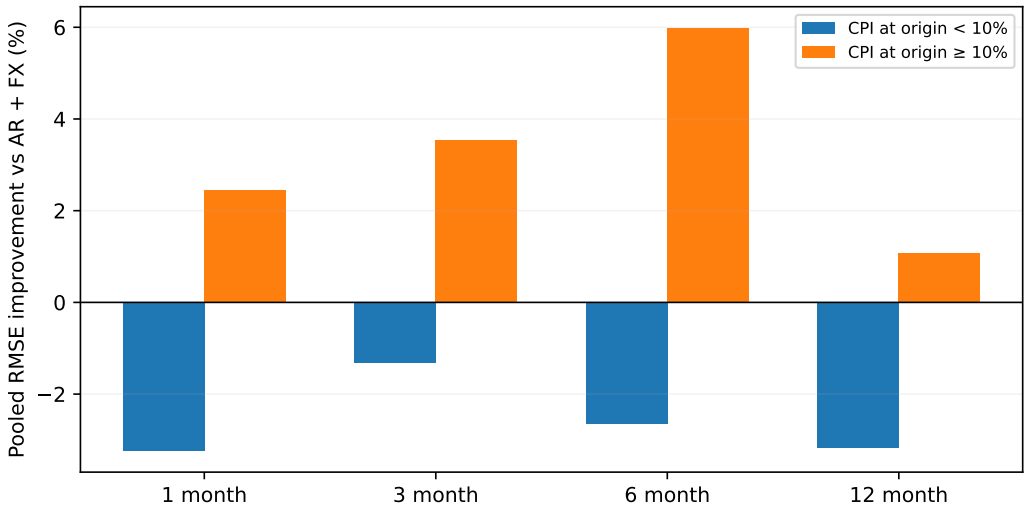


Figure 6: Forecast gains by the inflation regime observed at forecast origin

Notes. The origin-specific 10% classification does not use subsequent inflation, but the tests of regime differences in normalized squared losses are not statistically significant.

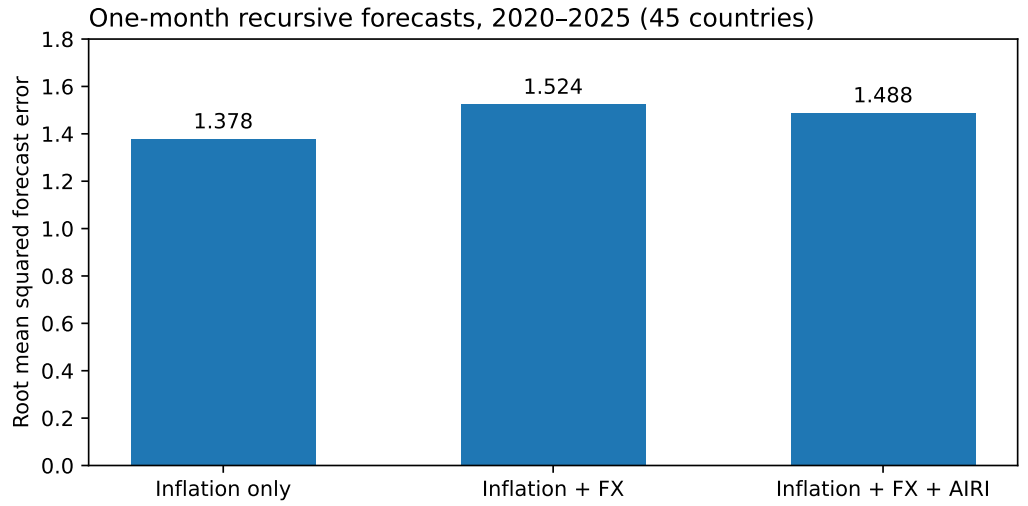


Figure 7: Pooled forecast RMSE across four horizons

Notes. Lower RMSE represents greater realized accuracy. Forecasting models are estimated country by country with expanding training samples.

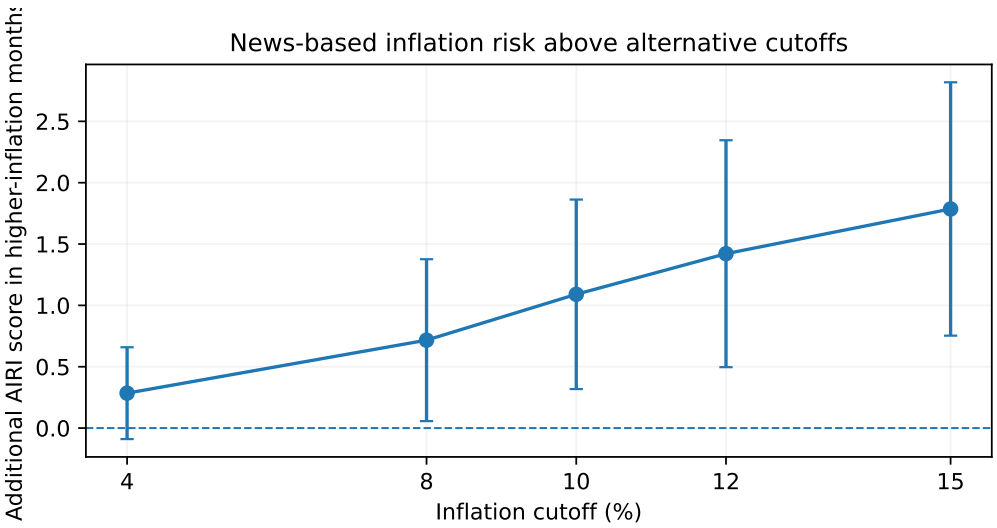


Figure 8: AIRI and the double-digit inflation environment

Notes. The graph summarizes the newspaper-risk regression. The indicator is based on preceding CPI inflation; the conditional association is not a causal effect of household attention.